

Static Block V/S Instance Initializer Block (Comparison)

Property	Static Block	IIB
Key word	static { }	none { }
Executes	Once	Every Object
Runs	When Class Loads	When object is created
Before	main()	Constructor
Used for	Class-level setup	Object-level setup

* package → folder containing similar classes & interfaces.

Industry Standard to create packages:

accenture ↔ icici ↔ login

com.company-name.project-name.module-name.filename.java;

com.accenture.icici.login.Login.java;

com
|
acc
|
icici
|
login login.java

* Square Root of a number using Binary Search
(Integral Part) $x \times x = n$ $(\log n)$

int sqrt(int n) {

{ $3 \times 3 = 9$
s = m+1

0 - 3 4 - 7

$\frac{11}{2} = 5$

$5 \times 5 = 25$
s = m+1

ans = 0

7 x 7 = 49

0 - 7

ans = mid (7)
s = mid+1

8 - 14

$11 \times 11 = 121$
e = m-1

8 - 10

ans = 0

ans = mid (3)

ans = mid (5)

6 - 7

$\frac{13}{2} = 6$

$6 \times 6 = 36$

0 - 64

m = 32

e = m-1

0 - 31

m = 15

e = m-1

0 - 14

m = 7

8 - 8

0 - 17

m = $\frac{17}{2} = 8$

e = m-1

0 - 7

m = $\frac{7}{2} = 3$

0 - 36

s = $\frac{s+e}{2} = 18$

e = m-1

18 x 18 = 324

0 - 17

m = $\frac{17}{2} = 8$

e = m-1

0 - 7

m = $\frac{7}{2} = 3$

0 - 64

m = 32

e = m-1

0 - 31

m = 15

e = m-1

0 - 14

m = 7

8 - 8

0 - 17

m = $\frac{17}{2} = 8$

e = m-1

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0 - 36

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